

SE LAB-3 DELIVERABLES:

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SECTION:G

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Scenario Review

- Student submits a project ZIP
- System queues the project
- Test Execution Environment executes tests
- System generates rubric scores
- Projects can be assigned to peer reviewers
- Users can view evaluation results
- System must handle up to 100 concurrent submissions
- Only authorized users can access submissions/results

Main challenges

Performance

- Handle up to 100 concurrent project submissions.
- Avoid losing queue items.
- Stay within the 1.5 GB memory requirement.

Security

- Student submissions and evaluation results must only be accessible to authorized users.

Scalability/Reliability

- Multiple projects may need to be evaluated simultaneously.
- Queued projects should not be lost

Architectural Style Analysis

Architecture	Advantages for our system	Disadvantages
Layered	Clear separation of UI, business logic and data; easier to develop and maintain	Less flexible for independently scaling evaluation components
Microservices	Evaluation, queue, scoring and review services can scale independently; failure isolation	More complex communication and deployment
Client-Server	Centralized control and relatively simple deployment	Central server can become a bottleneck when many projects are submitted

Select Architecture

Microservices Architecture

Microservices fits because evaluation-related functions can be separated into independently managed services.

Reason 1 — Scalability

The evaluation workload can be separated from submission and results management, allowing evaluation-related services to handle increased workload independently.

Reason 2 — Fault isolation

If the test execution/evaluation service encounters a failure, other functions such as viewing previous results do not necessarily need to stop.

Security advantage

Authorization can be enforced at service/API boundaries so that only authorized users can access student submissions and evaluation results.

Performance benefit

Evaluation workloads can be processed independently, allowing multiple submissions to be handled without forcing every operation through one central application process.

Identify EXACTLY 5 Components

1. Submission Management Component

Handles:

- Student project upload
- Submission validation
- Submission information

2. Queue Management Component

Handles:

- Queuing submitted projects
- Maintaining evaluation order
- Managing queued evaluation jobs

3. Evaluation/Test Execution Component

Handles:

- Running the configured test suite
- Recording test results

4. Rubric & Peer Review Component

Handles:

- Generating itemized rubric scores
- Assigning eligible peer reviewers

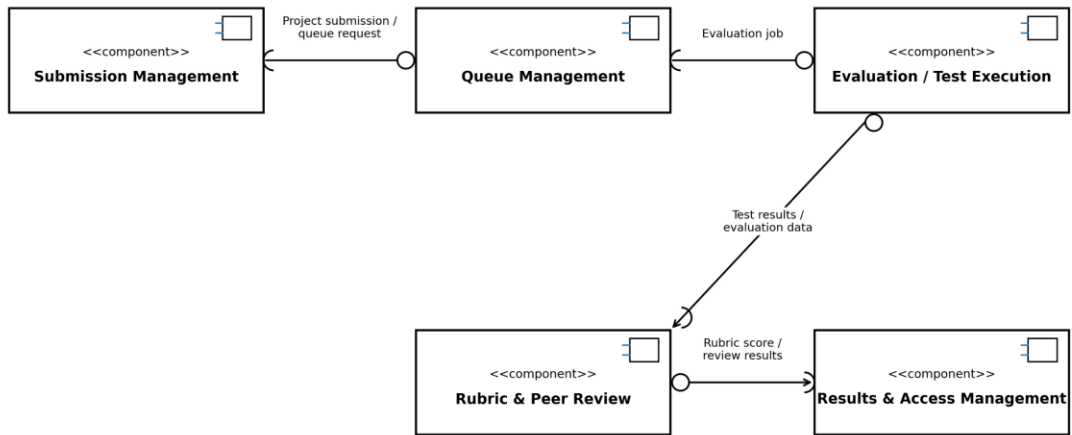
5. Results & Access Management Component

Handles:

- Storing/providing evaluation results
- Allowing authorized users to view results
- Enforcing access restrictions

Automated Rubric Assignment Evaluator

UML Component Diagram — Microservices Architecture



○ Provided interface ◐ Required interface (socket)

Solid arrows indicate the direction of the service/data interaction.